

Vedant Desai

(248) 704 – 4852 | vedantde@umich.edu | github.com/Verdent06 | linkedin.com/in/vedantde06

EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ~2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ~800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

Vylet | *Python, LangGraph, PostgreSQL, Docker, Redis, Celery*

May 2026 – Present

Founder — Lead-sourcing SaaS; \$1,500 MRR across three paying clients

vyletdata.com

- Launched Vylet, automating a ~30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Shipped a geography-first discovery mode decoupled from a pre-set vertical, surfacing ~30 leads/month in niche verticals — custard shops, bowling alleys — structurally invisible to vertical-first discovery.

Lyndbrook Capital

Feb 2026 – Apr 2026

Data Engineering Consultant

Remote

- Contracted pre-LOI to build acquisition intelligence for a boutique search fund targeting water utility operators; aggregated EPA ECHO and MassGIS regulatory data into a unified PWSID entity database and delivered 800+ validated Day-1 acquisition targets within the engagement window.
- Automated off-market deal sourcing by aggregating 2,500+ legal entities from EPA compliance databases and cross-referencing Google Maps API data to surface unlisted acquisition targets, eliminating 15 hours of manual prospecting per week for the fund's Principal.

PROJECTS

SignalWeaver | *FastAPI, PostgreSQL, pgvector, React, TypeScript, Docker* github.com/Verdent06/SignalWeaver

- Containerized the stack with Docker Compose (API + Postgres/pgvector + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).
- Implemented semantic search over stored financial news with pgvector cosine similarity (768-d MPNet embeddings), hitting 49ms p50 / 99ms p99 over 90 queries returning top-5 or top-10 matches.
- Built a React/TypeScript dashboard for composite scores and fundamental/sentiment/macro breakdowns; used it to analyze 90 distinct tickers with scores persisted to Postgres for history views.

Granular Synthesizer Plugin | *C++, JUCE, Lock-free Concurrency* github.com/Verdent06/granular-synth

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.

TECHNICAL SKILLS

Languages Python, TypeScript, C++, SQL, HTML/CSS
Frameworks React, FastAPI, Flask, JUCE
Databases PostgreSQL, Redis, pgvector
Cloud & Infrastructure AWS (EC2), Docker, Celery, Git, GitHub Actions